

Mathematics for Machine Learning

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*If the outcomes of any group of the trials do not affect the probabilities of the outcomes of any other trial, then the trials are called **independent trials**.*

Bernoulli Trials

Suppose there are only two possible outcomes in the sample space of each trial classified as "Success" and "Failure". Such trials are called **Bernoulli trials**. If moreover, the probability p of Success remains the same for each trial throughout the entire experiment, then such trials are called **repeated** Bernoulli trials.

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Binomial Experiment

Let n be some fixed positive integer. An experiment consisting of n repeated Bernoulli trials is called a **Binomial Experiment**.

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- Tossing a fair coin 10 times:
Success = Heads, Failure = Tails
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- Tossing a fair coin 10 times:
Success = Heads, Failure = Tails
 $p = 0.5$
- Rolling a fair die 5 times:
Success = opened number is even,
Failure = the other numbers
 $p = \frac{3}{6} = \frac{1}{2}$.

Binomial Probabilities

Suppose a Binomial Experiment has n trials. Let X be the number of successes in a sequence of n trials. Then for $0 \leq k \leq n$,

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Example

We are tossing a fair coin 100 times. The success is "Heads".

- 1) What is the probability of being successful exactly 5 times?*
- 2) What is the probability that we will be successful at least once?*

Example

An examine consists of 16 multiple-choice questions. Each question provides 4 answers, where only 1 is correct. A student, who is absolutely unfamiliar with the subject, chooses 16 answers randomly.

- 1) What is the probability that the student will answer correctly at least once?*
- 2) The exam is passed if at least 3 questions are answered correctly. What is the probability that the student above will pass the exam?*

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Example

A communication system consists of 5 components, each of which will, independently, function with probability p . The total system will be able to operate effectively if at least one-half of its components function. Find the probability that a 5-component system will be effective.

Multinomial Probabilities

Suppose an experiment has n independent trials and r possible outcomes:

$$A_1, A_2, \dots, A_r.$$

If A is the event that in n trials the event A_i will occur k_i times for all $i = 1, 2, \dots, r$ and $k_1 + k_2 + \dots + k_r = n$, then

$$\mathbb{P}(A) = \frac{n!}{k_1! k_2! \dots k_r!} p_1^{k_1} p_2^{k_2} \dots p_r^{k_r},$$

where $\mathbb{P}(A_i) = p_i$, for all $i = 1, 2, \dots, r$.

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Proposition

If X is a discrete random variable that takes on one of the values $x_i, i \geq 1$, with respective probabilities $p(x_i)$, then,

$$\mathbb{E}[X] = \sum_n x_n p(x_n).$$

moreover, for any real-valued function g ,

$$\mathbb{E}[g(X)] = \sum_n g(x_n)p(x_n).$$

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Proposition

If X is a continuous random variable, then for any real-valued continuous function g ,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx.$$

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In words, the expected value of discrete random variable X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it.

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Example

Find $E[X]$ where X is the outcome when we roll a fair die.

Properties

- 1 If $X(\omega) = I_A(\omega)$ is the indicator function for the event A , then $E[X] = \mathbb{P}(A)$.

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- 3 $E[X + Y] = E[X] + E[Y]$
- 4 $E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$
- 5 If $X \geq Y$, then $E[X] \geq E[Y]$.

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Random variables X, Y are called independent, if for all $x, y \in \mathbb{R}$ holds

$$\mathbb{P}(X \leq x \cap Y \leq y) = F_X(x) F_Y(y).$$

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If X and Y are independent RVs, then $E[XY] = E[X] E[Y]$.

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- 2 $\text{Var}(X) \geq 0$
- 3 If X and Y are independent random variables, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Example

Calculate $\text{Var}(X)$ if X represents the outcome when a fair die is rolled.

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Properties of Binomial Random Variables

If X is a binomial RV with parameters n and p , then

$$\mathbb{E}[X] = np, \text{Var}(X) = np(1 - p).$$

Expected Value and Variance

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$$\begin{aligned}\mathbb{E}[X^2] - \mathbb{E}[X] &= \sum_{k=1}^n k(k-1) C_n^k p^k (1-p)^{n-k} \\ &= n(n-1)p^2 \sum_{k=2}^n C_{n-2}^{k-2} p^{k-2} (1-p)^{n-k} = n(n-1)p^2.\end{aligned}$$

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$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = np + n(n-1)p^2 - (np)^2 = np(1-p).$$

Expected Value and Variance

Markov's Inequality

If X is a random variable, then for any value $\varepsilon > 0$

$$\mathbb{P}(|X| \geq \varepsilon) \leq \frac{E[|X|]}{\varepsilon}.$$

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Theorem

If X is a random variable, then for any value $\varepsilon > 0$ and $r > 0$

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Chebyshev's inequality

If X is a random variable, then for any value $\varepsilon > 0$

$$\mathbb{P}(|X - E[X]| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

Definition

A random variable X that takes on one of the values $0, 1, 2, \dots$ is said to be a Poisson random variable with parameter $\lambda > 0$ if the PMF of X is representable as

$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

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Poisson Distribution

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The fact that X is a Poisson RV with parameter λ is denoted as $X \sim Po(\lambda)$ or $X \sim Poiss(\lambda)$.

The Poisson random variable may be used as an approximation for a binomial random variable with parameters (n, p) when n is large and p is small enough so that $n \cdot p$ is of moderate size.

If $X \sim B(n, p)$, then $P(X = k) \approx \frac{\lambda^k}{k!} e^{-\lambda}$, where $\lambda = np$.

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Example

Assume that the number of university professor in Armenia is 400 and the probability of reaching age 100 is 0.005. Then let's assume that there are only $n = 400$ professors. Find the probability that at least 3 professors will reach age 100.

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Example

Let X be a Poisson random variable with parameter λ . Prove that

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

Expected Value and Variance

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